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(54) **PUMP-BASED SYSTEM FOR WASTEWATER TREATMENT THROUGH MICROORGANISM BIOCHEMICAL PATHWAY OPTIMIZATION**

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C02F 1/36 (2013.01); C02F 3/006 (2013.01);
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See application file for complete search history.

(56)

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C02F 3/00 (2023.01)

C02F 3/12 (2023.01)

C02F 3/20 (2023.01)

C02F 3/28 (2023.01)

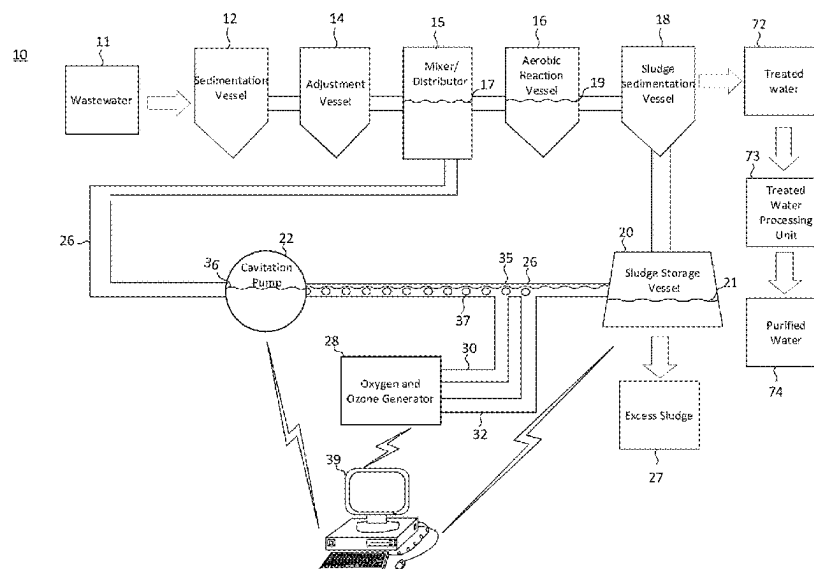
(52) **U.S. Cl.**

CPC **C02F 3/26** (2013.01); **B01F 31/80** (2022.01); **C02F 3/1221** (2013.01); **C02F 3/20** (2013.01); **C02F 3/28** (2013.01); **B01F**

ABSTRACT

Increased control and efficiency over the wastewater purification can be achieved through creating conditions that allow the operator to selectively prioritize the digestive function of microorganism in the activated sludge. The gas-dispersion return sludge is created using pure oxygen or oxygen containing trace amounts of ozone as a reactive gas, which is blended with return sludge to create a mixture of gas and liquid, which is passed through an atomizer or a cavitation pump to instantly render the reactive gas to an ultra-fine bubble state. At least a portion of the ultra-fine bubbles dissolve within the gas-dispersion return sludge, activating the dormant microorganisms. Due to a complete or an almost complete absence of biodegradable material in the gas-dispersion return sludge, the microorganism prioritize their digestive function, and when exposed to biodegradable pollutants present in wastewater, digest the pollutants using biochemical pathways different from the ones used in nature.

20 Claims, 5 Drawing Sheets



Related U.S. Application Data

continuation of application No. 17/245,631, filed on Apr. 30, 2021, now Pat. No. 11,724,948, which is a continuation of application No. 16/866,434, filed on May 4, 2020, now Pat. No. 11,046,603, which is a continuation-in-part of application No. 16/665,533, filed on Oct. 28, 2019, now Pat. No. 10,689,279, which is a continuation of application No. 16/517,637, filed on Jul. 21, 2019, now Pat. No. 10,457,582, which is a continuation-in-part of application No. 16/236,190, filed on Dec. 28, 2018, now Pat. No. 10,435,319, which is a continuation of application No. 15/956,631, filed on Apr. 18, 2018, now Pat. No. 10,167,214.

Fig. 1A.

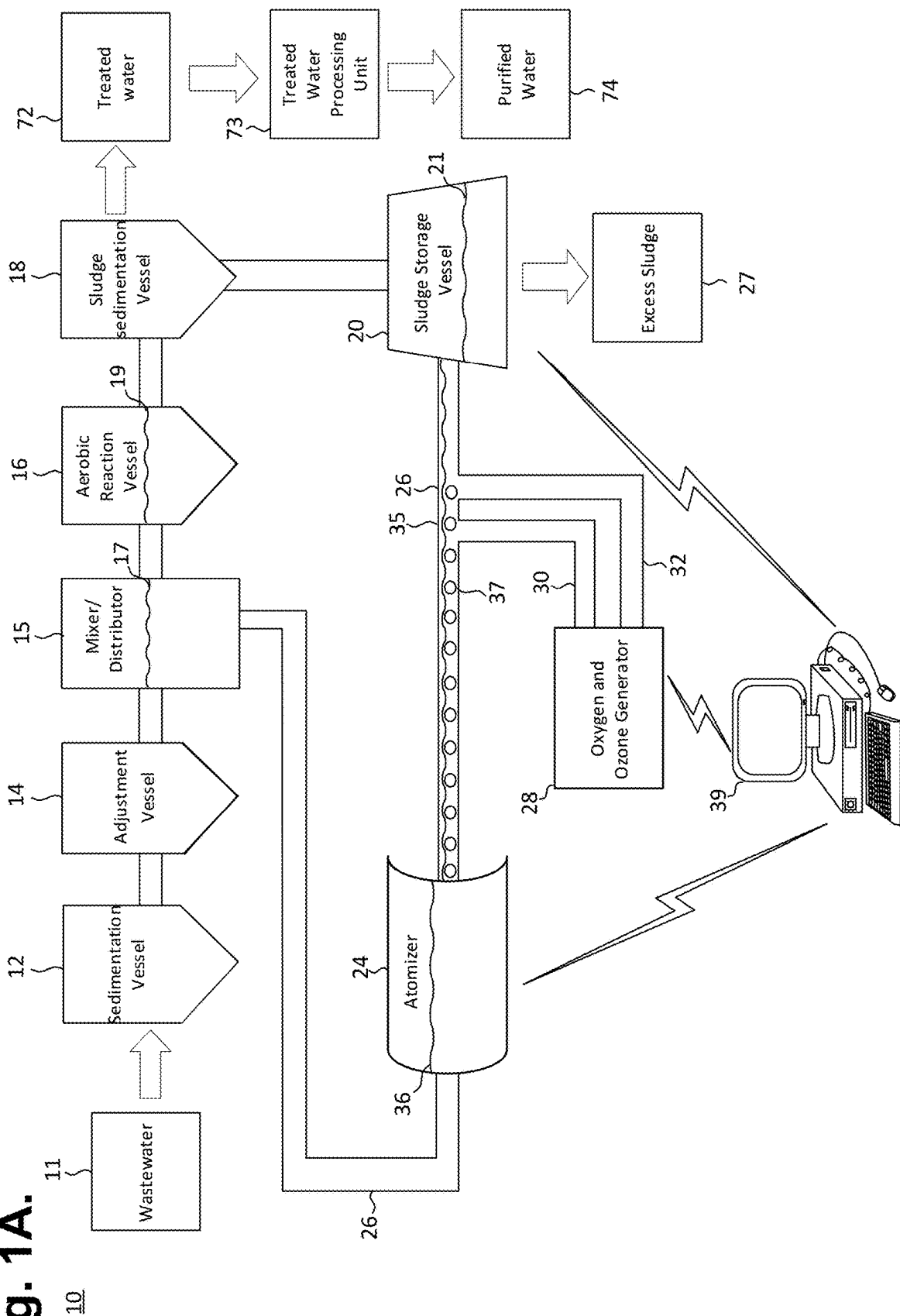


Fig. 1B.

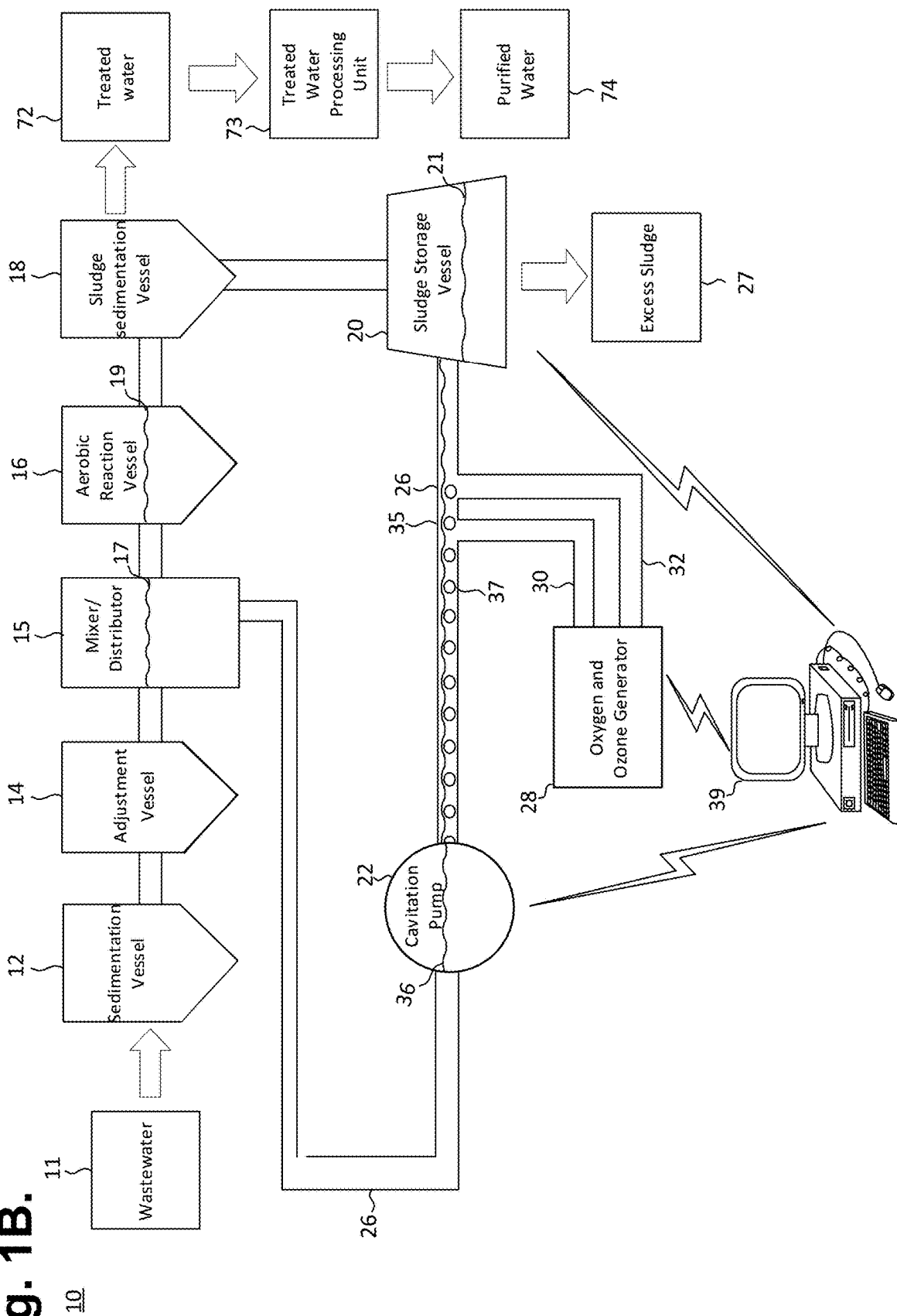


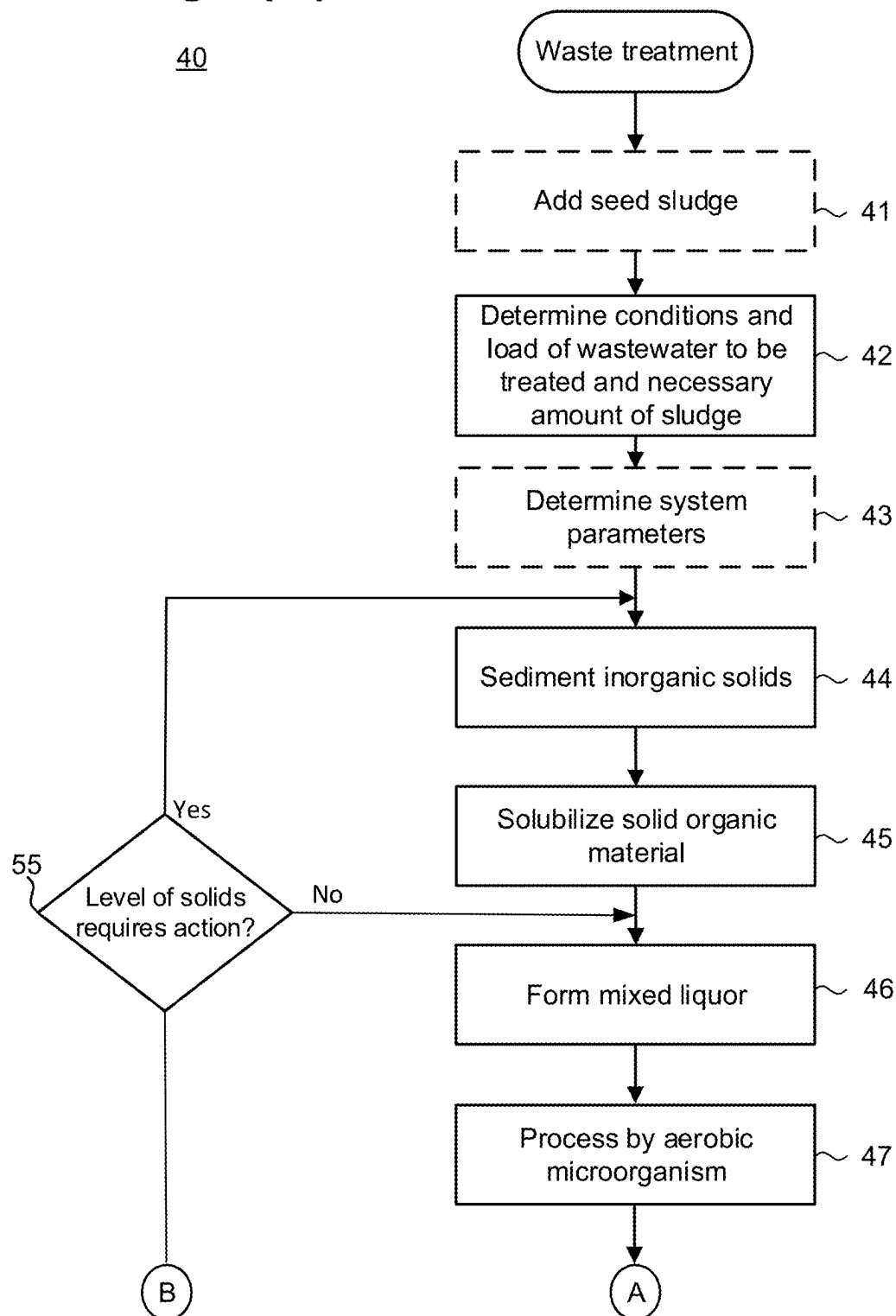
Fig.2(A).

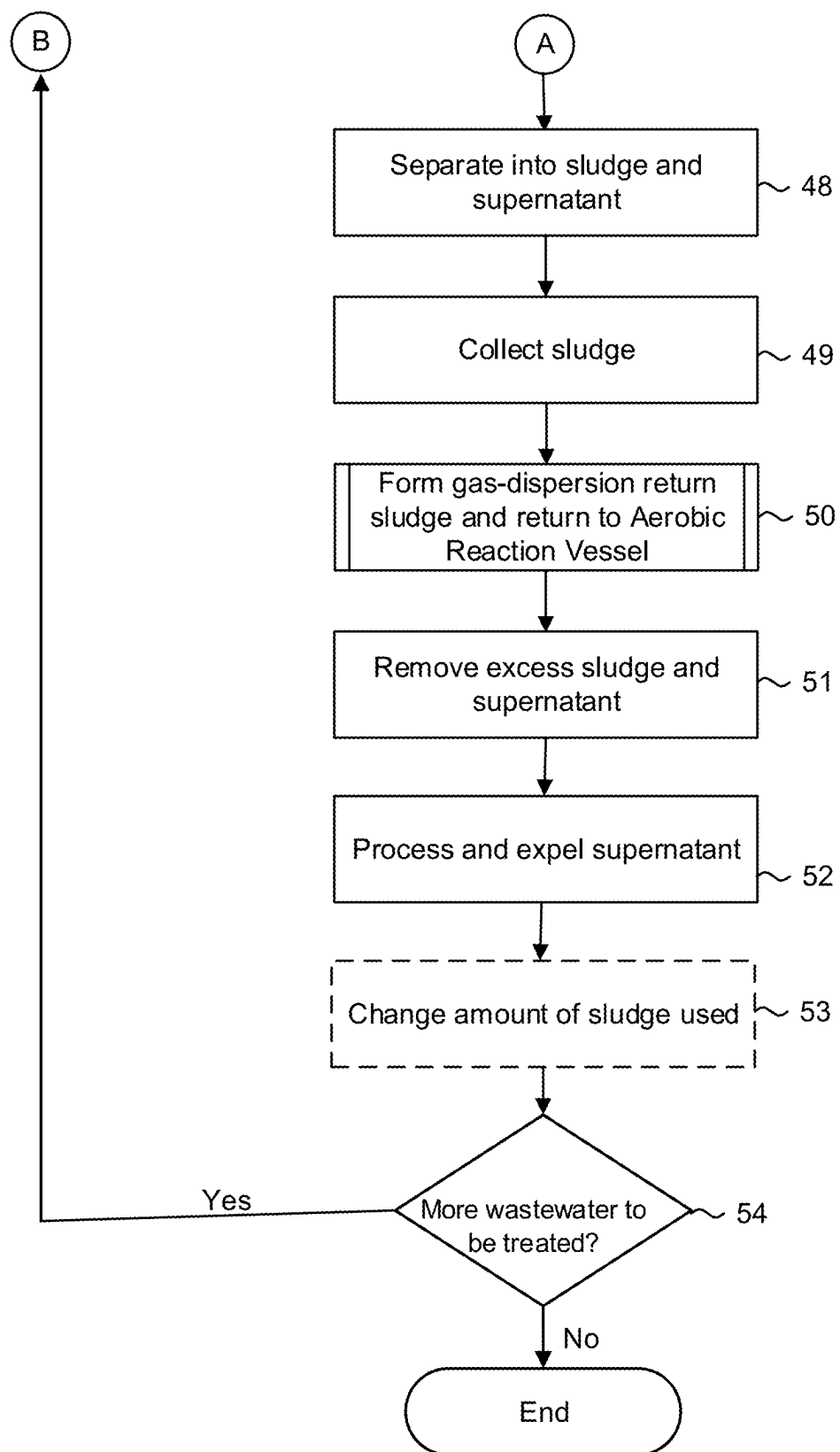
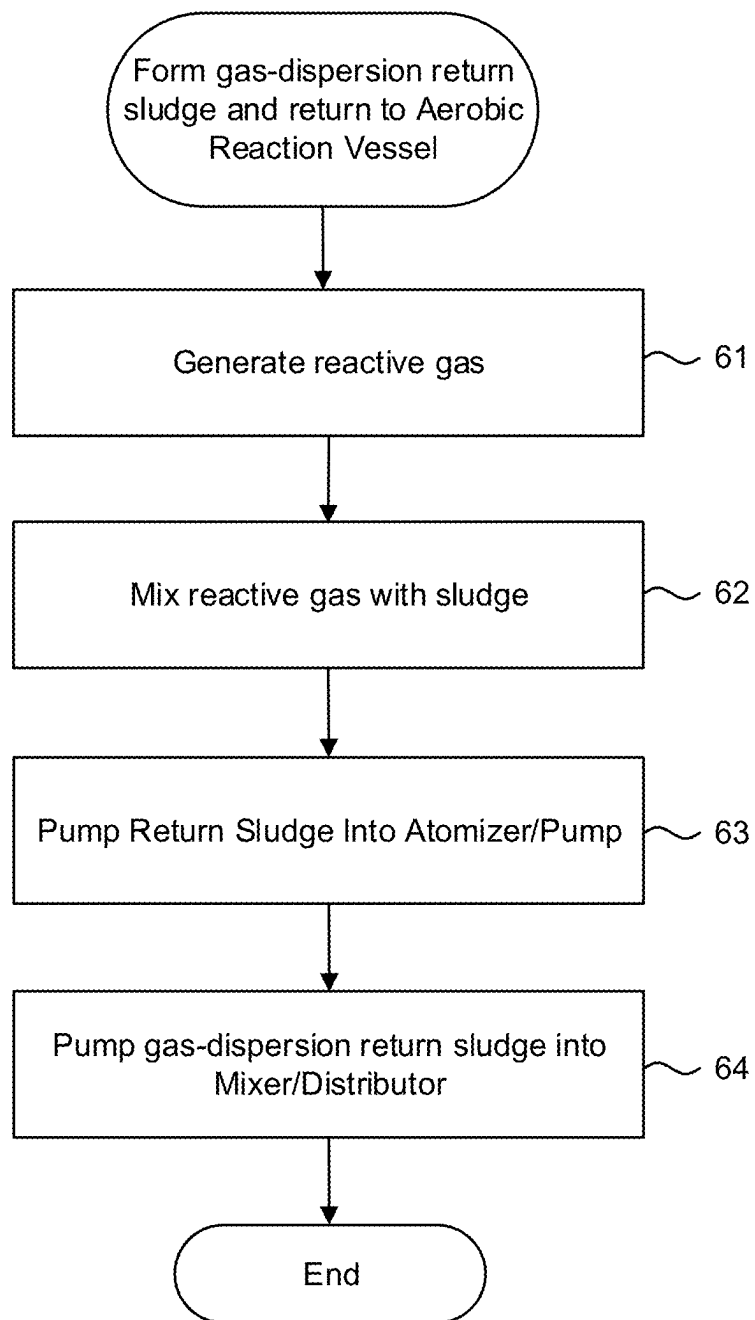
Fig. 2(B).

Fig. 3.60

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PUMP-BASED SYSTEM FOR WASTEWATER TREATMENT THROUGH MICROORGANISM BIOCHEMICAL PATHWAY OPTIMIZATION

FIELD

The present invention relates in general to wastewater purification, and in particular, to a pump-based system for wastewater treatment through microorganism biochemical pathway optimization.

BACKGROUND

The activated sludge method is employed widely today for the purification of wastewater. The activated sludge method is a biochemical treatment and oxidation process which employs microorganisms and oxygen to immobilize organic pollutant substances which are dissolved in wastewater into activated sludge utilizing the reproductive function of the sludge, and then utilizes the digestive function of the sludge to break down a portion of the organic pollutants into water (H_2O) and carbon dioxide gas (CO_2) for removal. Other pollutants present in the wastewater, such as ammonia, can be similarly broken down into water and other byproducts.

The typical activated sludge wastewater treatment techniques have over a century of history and many challenges are associated with such traditional techniques. For instance, the biochemical cleansing of organic pollutant substances depends largely on the quantity of microorganisms (return sludge), the density of the microorganisms, and the degree of their activity. However, to increase the quantity of microorganisms, their density, and their activity, increasing the supply of dissolved oxygen, which is essential to the microorganisms, is necessary. Without adequate supply of dissolved oxygen, the wastewater treatment may not be effective.

When the activated sludge method is employed under natural environmental conditions, namely, at 20° C. under standard pressure, 1DO (mg/L) of dissolved oxygen is required to purify 1 BOD (biochemical oxygen demand in mg/L) of organic pollutants in a five-day period. Similarly, 1DO (mg/L) of dissolved oxygen is required to purify 1COD (chemical oxygen demand in mg/L) of organic pollutants at 20° C. under standard pressure in a 30-minute to two-hour period. Therefore, under standard environmental conditions, the purification processing performance of the standard activated sludge method does not exceed 1BOD per 1DO, and in the same way, 1DO is required to purify 1COD. In other words, to purify either 1BOD of pollutant or 1COD of pollutant, 1DO of dissolved oxygen is required. As for the time required, 1 BOD of pollutant require five days and 1COD of pollutant requires 30 minutes to two hours.

While many enhancements and improvements have been proposed to traditional activated sludge-based wastewater treatment, most of them presume conditions which exist naturally in the environment. To date, no innovative technology or method that brings about a revolutionary improvement in performance has been proposed.

The activated sludge method employs microbes and oxygen to effect a biochemical treatment and oxidation, isolating organic pollutants in the wastewater in the form of activated sludge, so that a portion of the organic pollutants can be broken down, most commonly, to water (H_2O) and carbon dioxide gas (CO_2), for removal. For this reason, the biochemical purification of the organic pollutants depends

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greatly upon the quantity of return sludge (microbe flora), the density of the microbe flora, and the degree to which the microbe flora is activated.

One enhancement to traditional activated sludge-based wastewater treatment is known as "preliminary aeration." When preliminary aeration is used, the return sludge is aerated in advance, and the return sludge (microbe flora) thus activated is supplied to an aeration vessel. However, the capacity enhancement from preliminary aeration is limited to about 30%, and due to this low improvement ratio the cost of aeration is immense. The additional cost of aeration is roughly 100%, so for a 30% improvement in performance the cost is doubled, which is clearly not cost-effective.

Similarly, another technique used today is long-term continuous aeration bubbling technology, in which the wastewater to be purified and a return activated sludge are combined in an aeration basin into a mixed liquor. Air is provided through a blower into the aeration basin. Bubbles of about 1 mm are produced, aerating the mixed liquor so that the air is dissolved into the wastewater, providing oxygen for aerobic microorganisms and activating them so they can break down organic solids in the wastewater more efficiently. However, as oxygen is not easily soluble, even with the bubbling, the achieved concentration of dissolved oxygen is not high enough to bring about a large increase in microorganisms, generally being 2-4 mg/l, a level similar to what is observed in nature, such as in rivers and lakes. While a greater number of microorganisms can be provided by increasing the amount of return sludge inserted into the aeration basin, to be effective, the increase would have to be accompanied by increasing the supply of available oxygen, which may not be possible without physically changing the existing setup. Further, currently any changes to the existing set-up, including the size and production capacity of any source of oxygen, would likely involve guesswork as to what changes would be sufficient to provide the required oxygen, with no precise relationship between the amount of oxygen provided and the amount of contaminants removed being known.

Likewise, U.S. Pat. No. 7,105,092, issued Sep. 12, 2006, to Kousuke Chiba ("092 patent"), the disclosure of which is incorporated by reference, discloses a sewage treatment process by which activated-sludge is produced using a line atomizing treatment. Wastewater is introduced into the treatment line. The wastewater passes through the adjustment vessel and the sedimentation vessel where inorganic pollutant substances are removed. Subsequently, the wastewater enters the anaerobic reaction vessel where the wastewater is acted upon by anaerobic microorganisms. Subsequently, the wastewater enters the aerobic reactive vessel where organic matter within the wastewater is converted into activated sludge by the action of aerobic microorganisms. After the conversion process in the aerobic reaction vessel, the treated wastewater solution which has had the dissolved organic matter converted into activated sludge is sent together with the activated sludge to the sludge sedimentation vessel, and the supernatant water is expelled from the wastewater treatment system. The supernatant water may also be subjected to advanced treatment for further purification.

The '092 patent further discloses that a portion of the activated sludge which has settled in the sludge sedimentation vessel passes through the sludge intake pipe and is supplied respectively as return sludge to the adjustment vessel, sedimentation vessel, anaerobic reactive vessel, aerobic reactive vessel, and sludge sedimentation vessel to effect multiple functionality for each of those vessels, and to enhance the treatment capacity of the wastewater system

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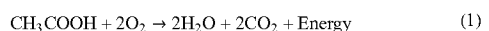
while allowing the remainder of the activated sludge to undergo separate treatment as excess sludge. However, each vessel has an original function and role, and in many cases, adding activated return sludge which holds large quantities of reactive gases (oxygen or oxygen with trace amounts of ozone) may interfere with those functions or roles, thus decreasing the effectiveness of wastewater treatment.

Further, the cleansing of wastewater depends fundamentally on the activity of microorganisms (activated sludge), and is thus saddled with the problem of the formation of excess sludge due to the excessive reproduction of these microorganisms, and technology to control this excess has not yet adequately been realized. In other words, the microorganisms which are involved in the cleansing of wastewater are constantly reproducing themselves and then perishing due to self-oxidization, hence controlling and managing the amount of sludge produced and the amount destroyed is extremely difficult, and the lack of this control and management is considered the critical problem of the activated sludge method. As a result, the large quantities of excess sludge that form are concentrated, transported and incinerated or buried in landfills, causing massive processing costs for the removal of excess sludge and emissions problems from the release of carbon dioxide during incineration.

In the activated sludge method, the activated sludge, that is, the microbe flora, which purifies the organic and other pollutants in the wastewater, can be considered to be purifying primarily through the following functions: the reproductive function, where the microbe flora absorbs organic matter as food, the flora grows and multiplies via asexual reproduction, and the absorbed matter is isolated in the form of a clump of microbes; and the digestive function, where the microbe flora absorbs organic and other matter as food, and digests the food to release energy which it uses to stay alive and carry out its life processes. To efficiently purify the organic and other chemical matter in wastewater, there is an essential need for the sludge be activated, but under normal environmental conditions, the entire microbe flora is activated and controlling or managing the digestive and reproductive functions separately is impossible. Under normal conditions, the reproductive function is liable to increase, creating large quantities of excess sludge. To purify organic and other matter efficiently, bringing these two functions into balance is necessary. Accordingly, there is a need to be able to control and manage both digestive and reproductive functions separately.

In the activated sludge method, the wastewater purification capacity is fundamentally dependent upon the activity of the microbe flora (activated sludge). For this reason, while having the microbes activated is indispensable for increasing the wastewater purification capacity, wastewater purification techniques that simply activate in a directionless fashion result in excess reproduction, bringing about the problems of excess sludge which are among the most fundamental issues with the activated sludge method.

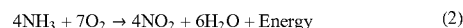
Further, even if the digestive function is exhibited by the microorganisms to a desired extent, the digestive function can be exhibited via multiple biochemical pathways, not all of which are desirable. For example, under normal conditions (conditions encountered in nature) digestion of acetic acid by microorganisms proceeds in accordance with the following equation:



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Structurally similar organic pollutants are similarly processed. Such reaction consumes two molecules of diatomic oxygen for every molecule of acetic acid being digested, quickly depleting the amount of oxygen necessary to sustain the reaction. Further, carbon dioxide is a greenhouse gas, and treating activated sludge primarily through such a reaction contributes to global warming, being undesirable when employed on industrial scale.

Likewise, exhibiting the digestive function under normal conditions on other kinds of pollutants can create undesirable byproducts. For instance, under normal conditions, ammonia is digested by microorganisms in accordance with the following equation:



NO_2 , nitrous oxide, is also a greenhouse gas, and wastewater treatment utilizing this reaction on an industrial scale for an extended period of time can make a noticeable contribution to global warming. Further, for every four ammonia molecules being digested, the reaction consumes seven molecules of diatomic oxygen, quickly depleting the amount of oxygen available to sustain the wastewater treatment and requiring additional efforts to maintain the necessary oxygen concentration in the wastewater being treated.

However, there is a lack of technology available today that can handle these issues. Specifically, the microbe flora goes through a constant cycle of growing through reproduction but then extinction of the microbes by digestion, and the effective separate control and management of the growth and extinction of microbe flora as caused by these two functions is considered an extremely difficult problem. Further, the current technology is unable to direct the digestive function via pathways that do not produce undesirable byproducts or consume excessive amounts of oxygen.

Accordingly, there is a need for a way to control the purifying function of the microbe flora of activated sludge such that the purifying effect can be utilized technologically and industrially.

SUMMARY

The system and method described below, firstly, enable separate control and management of the reproductive function and the digestive function of the microbes which are the key factors in the activated sludge-based wastewater treatment, thereby limiting the production of excess sludge. Secondly, whereas in the activated sludge method 1DO is defined as the unit of purifying 1BOD or 1COD, the system and method described below, by limiting the reproductive function and enhancing the digestive function, allows the purification of a far greater quantity of organic matter than in the traditional method where no more than 1BOD or 1COD can be purified with 1DO of dissolved oxygen. Thirdly, the system and method described below allow the individual control and management of the reproductive and digestive function in the activated sludge-based wastewater treatments. In addition, the system and method described below allow to direct the digestive function via biochemical pathways that do not produce undesirable byproducts, such as greenhouse gases and do not consume as much oxygen as under normal conditions. Fourthly, the system and method described below can be easily and inexpensively fitted to the activated sludge-based wastewater treatment facilities currently in use throughout the world.

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The system and method described below focuses on the digestive function of microbe flora which absorbs organic matter as food and breaks the organic matter down for sustenance and life energy, and causes the microbe flora to exhibit their digestive function more than their reproductive function by creating or providing a near-starvation state environment which is not dependent upon the respective quantities of microbes and food. In particular, the system and method described below create conditions that do not exist in nature to activate the microbe flora to a high degree, providing an extreme starvation environment, spurring the microbes to exhibit the digestive function over the reproductive function, such that the quantity of organic matter which can be digested with 1DO is far greater than the 1BOD or 1COD which was the maximum attainable in the traditional activated sludge-based purification. Thus, by creating or providing conditions which do not exist in nature, the system and method described below stimulate strongly and to a high degree the activated sludge (microbe flora) which purify the organic matter and other matter in wastewater, separately controlling and maintaining the reproductive function and the digestive function which the microbes possess, thereby yielding an effect which is revolutionary and unimaginable in nature. Further, under these conditions that do not exist in nature, the digestive function utilizes biochemical pathways that consume less (if any) oxygen than is found under conditions and, in addition to water, create byproducts that are inert, such as amorphous carbon or diatomic nitrogen (N_2).

Some of the features of the system and method described below can include: using pure oxygen (O_2) or oxygen with a trace amount of ozone (O_3) (e.g. less than 0.5 mg/L of sludge) as a reactive gas; super-saturating the return sludge with pure oxygen (at least DO 10 mg/L); dissolving trace amounts of ozone (less than 0.5 mg/L) in the return sludge; supplying to the aerobic reaction vessel return sludge with pure oxygen or the oxygen/ozone mixture in the quantity equal to at least 10% of the wastewater to be treated; and activating the entire microbe flora (all of the return activated sludge).

By providing these features, individually or in a group (including providing all of the conditions), the microbe flora is activated. The activation takes place when the organic matter which serves as food to the microbes is cut off. When the activated microbe flora reaches an extreme state near the point of starvation, the microbes begin to absorb organic matter to the limit of their capacity, regardless of the quantity of food or the quantity of microbes, and they prioritize their digestive function via pathways that lead to a greatest amount of energy and a least amount of waste, deepening the degree of wastewater purification. The extreme state near the point of starvation here refers to a state in which the microbes, having been activated to a high degree, enter a state of extreme near-starvation, pushing them to the point where they become desperate for food, after which they suddenly encounter food (in the form of a mixture of return sludge and wastewater). As a result, the reproductive function of the microbes becomes suppressed and the production of large quantities of excess sludge is prevented. Specifically, by forcing the microbes into a state of near-starvation, it is possible to control and monitor the functions which the microbes naturally possess. In particular, the use of the system and method below allows the operator to cause the microbes to prioritize digestion over reproduction, which suppresses the production of excess sludge.

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Further, because the microbes are activated strongly by the introduction of reactive gas into the Gas-Dispersion Return sludge, a high level of preliminary aeration can be observed, and wastewater treatment capacity is dramatically enhanced. As a result of the enhanced wastewater treatment capacity, use of a far smaller aerobic reaction vessel becomes possible, eliminating the high-cost, energy-wasting processes of bubbling and churning which are necessitated in the conventional activated-sludge based techniques because of the poor water solubility of oxygen. Further, due to the microbe flora being activated to a high degree under environmental conditions which do not exist in nature, wastewater purification can be achieved with a much greater efficiency and in a far shorter time than in the conventional activated sludge method in which organic matter is defined in terms of BODs and CODs.

The system and method described below can be implemented on any existing wastewater treatment facility easily and with very small investment, and can be retrofitted, and can provide industrial energy savings and economic benefit on a global scale. In particular, the system and method allow the operator to solve the problem of the huge expenditures on energy for inefficient bubbling of insoluble air used as reactive gas, as well as the requirement for gigantic aeration facilities, and the problem of the production of large amounts of excess sludge during wastewater purification can be solved simultaneously, with a potential to bring about a massive economic effect to the entire world. Further, as world population is rapidly concentrating in cities, this invention allows for the rebirth of the activated sludge-based as a low-cost, high-efficiency, low-energy-consuming urban infrastructure technology.

In one embodiment, a pump-based system for wastewater treatment through microorganism biochemical pathway optimization is provided. The system includes a gas generator configured to provide at least one reactive gas into a sludge that includes microorganisms having a digestive function capable of digesting at least one pollutant via at least two biochemical pathways, one of the pathways for digesting that pollutant producing inert byproduct and another one of the pathways producing a further byproduct other than the inert byproduct, wherein the sludge is substantially free of the pollutants; a controller interfaced to the gas generator and a pump and configured to receive user input comprising a desired degree of the digestion of the least one pollutant in wastewater via the digestive function and to determine an amount of a gas-dispersion return sludge used to form a mixed liquor based on the user input; and the pump configured to: form a gas-dispersion return sludge by rendering the at least one reactive gas into ultra-fine bubbles within the sludge, wherein a portion of the ultra-fine bubbles dissolves within the sludge; and provide under a control of the controller the determined amount of the gas-dispersion return sludge for combining with a wastewater that includes the pollutant to form the mixed liquor, wherein, upon encountering the pollutant within the wastewater, the activated microorganisms digest the pollutant by prioritizing the pathway that produces the inert byproduct over the pathway that produces the other byproduct.

In a further embodiment, a pump-based controlled system for wastewater treatment through microorganism biochemical pathway optimization is provided. The system includes a gas generator configured to provide at least one reactive gas into a sludge that includes microorganisms having a digestive function capable of digesting at least one pollutant via at least two biochemical pathways, one of the pathways for digesting that pollutant producing inert byproduct and

another one of the pathways producing a further byproduct other than the inert byproduct, wherein the sludge is substantially free of the pollutants; a controller interfaced to the gas generator and a pump and configured to receive user input comprising a desired treatment time of the wastewater and to determine an amount of a gas-dispersion return sludge used to form a mixed liquor based on the user input; and the pump configured to: form a gas-dispersion return sludge by rendering the at least one reactive gas into ultra-fine bubbles within the sludge, wherein a portion of the ultra-fine bubbles dissolves within the sludge; and provide under a control of the controller the determined amount of the gas-dispersion return sludge for combining with a wastewater that includes the pollutant to form the mixed liquor, wherein, upon encountering the pollutant within the wastewater, the activated microorganisms digest the pollutant by prioritizing the pathway that produces the inert byproduct over the pathway that produces the other byproduct.

Still other embodiments of the present invention will become readily apparent to those skilled in the art from the following detailed description, wherein are described embodiments of the invention by way of illustrating the best mode contemplated for carrying out the invention. As will be realized, the invention is capable of other and different embodiments and its several details are capable of modifications in various obvious respects, all without departing from the spirit and the scope of the present invention. Accordingly, the drawings and detailed description are to be regarded as illustrative in nature and not as restrictive.

BRIEF DESCRIPTION OF THE DRAWINGS

FIGS. 1A-1B are block diagrams showing a system for wastewater treatment through controlled microorganism activation in accordance with two embodiments.

FIGS. 2A-2B are flow diagrams showing a method for wastewater treatment through controlled microorganism activation in accordance with one embodiment.

FIG. 3 is a flow diagram showing a routine for forming gas-dispersion return sludge and returning the gas-dispersion return sludge to the aerobic reaction vessel for use in the method of FIG. 2A-2B in accordance with one embodiment.

DETAILED DESCRIPTION

Traditionally, the relationship between the amount of oxygen provided to microorganisms and the amount of pollutants in wastewater that those microorganisms can break down using the oxygen has been thought to be linear. Thus, regardless of the overall concentration of oxygen, 1DO (mg/l) has been thought to be capable of facilitating the breakdown of 1BOD or 1COD of pollutants. As further described below, at high oxygen concentrations achievable through the use of the system and method described below, that relationship is no longer linear, and the achievable wastewater purification effect far exceeds the effect achievable with activated sludge-based purification when oxygen remains at a level seen in nature (2-4 mg/l). This effect increases the rate at which wastewater can be purified and decreases the cost of such purification.

Microorganisms, after being placed in an environment with very little food or a closed environment which causes near-starvation, can, upon contact with food, exhibit advantageously their digestive abilities over their reproductive abilities. The increase in the purification ability of the microorganisms in the presence of the high level of oxygen is achieved by making the microorganisms prioritize their

digestion function over their reproduction function. Microorganisms (also referred to as microbes below), such as bacteria and protozoa, prioritize their digestive function when the amount of available consumable material (such as organic pollutants) is low relative to the microorganism population. On the other hand, when the amount of the consumable material is high relative to the microorganism population, enough to satisfy the energy needs of the existing microorganism population as well as additional population, the reproductive function of the microorganisms is prioritized over the digestive function and the microorganisms asexually reproduce. In the description below, the microorganisms exhibiting their digestive and reproductive function can include aerobic microorganisms, as well as facultative anaerobic and opportunistic microorganisms. FIGS. 1A-1B are block diagrams showing a system 10 for wastewater treatment through controlled microorganism activation in accordance with two embodiments. The system 10 in the embodiment shown with reference to FIG. 1A includes a Sedimentation Vessel (also known as a sediment pool and a grit chamber) 12, an Adjustment Vessel 14, a Mixer/Distributor 15, one or more Aerobic Reaction Vessels (also known as an aeration vessel) 16, a Sludge Sedimentation Vessel 18, a Sludge Storage Vessel 20, an Atomizer 24, and a Treated Water Processing Unit 73. The system 10 shown with reference to FIG. 1B includes a Sedimentation Vessel (also known as a sediment pool and a grit chamber) 12, an Adjustment Vessel 14, a Mixer/Distributor 15, one or more Aerobic Reaction Vessels (also known as an aeration vessel) 16, a Sludge Sedimentation Vessel 18, a Sludge Storage Vessel 20, a Cavitation Pump 22, and a Treated Water Processing Unit 73. In a still further embodiment, the system 10 could include both a Cavitation Pump 22 and an Atomizer 24.

The Sludge Storage Vessel 20 and the one or more Aerobic Reaction Vessels 16 are connected by the Return Sludge Pipeway 26, constructed such that return sludge which has settled in the Sludge Storage Vessel 20 can be supplied to the Mixer/Distributor 15 and eventually to the one or more Aerobic Reaction Vessel 16. The Atomizer 24 or the Cavitation Pump 22 are positioned linearly along the Return Sludge Pipeway 26. Return Sludge 35 which travels through Return Sludge Pipeway 26 is therefore acted on by the Atomizer 24 (in the embodiment of the system 10 shown with reference to FIG. 1A) or the Cavitation Pump (in the embodiment of the system 10 shown with reference to FIG. 1B) and becomes gas-dispersion return sludge 36, which is in turn supplied to the Mixer/Distributor 15 and the one or more Aerobic Reaction Vessel 16 as gas-dispersion return sludge 36.

The system 10 further includes an Oxygen and Ozone Generator 28 that provides at least one reactive gas 37, oxygen with a possible addition of ozone, for addition to the return sludge 35. The Oxygen Supply Pipe 30 and the Ozone Supply Pipe 32 which are connected to the Oxygen and Ozone Generator 28 are connected to one or more pipes that is part of a Return Sludge Pipeway 26 and is in the upstream (intake) side of Atomizer 24 (in the embodiment of the system 10 shown with reference to FIG. 1A) or of the Cavitation Pump 22 (in the embodiment of the system 10 shown with reference to FIG. 1B).

For the Oxygen and Ozone Generator 28, the oxygen and ozone generator in U.S. Pat. No. 7,105,092 may be utilized. Other kinds of generators are possible. In one embodiment, the total amount of gas with produced by the Oxygen and Ozone Generator 28 includes no less than 90% of oxygen, with the concentration of ozone within the gas-dispersion

return sludge 36 not exceeding 0.5 mg/L and the concentration of oxygen being more than 10 mg/L.

The Return Sludge Pipeway 26 is connected only to the Mixer/Distributor 15 (and hence to the one or more Aerobic Reaction Vessels 16), and is therefore not connected to Sedimentation Vessel 12, Adjustment Vessel 14, and Sludge Sedimentation Vessel 18. The technological reason for this is discussed below.

As further described below beginning with reference to FIGS. 2A-2B, the Wastewater for Treatment (raw sewage) 11 enters Sedimentation Vessel 12 where grit and other inorganic solids settle and are separated out. From the Sedimentation Vessel 12, the Wastewater 11 flows into Adjustment Vessel 14 where the load and conditions of the inflowing raw sewage are adjusted, such as through automatic dilution, though other adjustments are possible, and organic solids present in the wastewater are solubilized by anaerobic microorganisms.

From the Adjustment Vessel 14, the Wastewater 11 flows into the Mixer/Distributor 15. The Mixer/Distributor 15 (also referred to as "Mixer" in the description below) receives wastewater 11 supplied from the Adjustment Vessel 14 and also receives gas-dispersion return sludge 36 supplied from via the Return Sludge Pipeway 26, mixes them together and supplies a mixture ("a mixed liquor 17") of gas-dispersion return sludge 36 and wastewater 11 to the one or more Aerobic Reaction Vessels 16. From there, aeration by bubbling using air as a reactive gas becomes unnecessary due to the reactive gas contained in gas-dispersion return sludge 36, which supplies dissolved oxygen (DO) to the Aerobic Reaction Vessel 16 (and initially to the Mixer 15). The Mixer 15 can include such parts as necessary for carrying out this function, such as a vessel for temporarily holding the mixed liquor 17, connections to the Aerobic Reaction Vessels 16, and one or more pumps for pumping the mixed liquor 17 to the one or more Aerobic Reaction Vessels 16. When there are multiple Aerobic Reaction Vessels 16, the Mixer/Distributor 15 can be managed appropriately to distribute to each of them in accordance with the treatment capacity of each. In an installation with only one Aerobic Reaction Vessel, the Mixer/Distributor 15 can be entirely omitted, with the wastewater 11 and the gas-dispersion return sludge 36 being pumped directly into the Aerobic Reaction Vessel 16, where they form the mixed liquor 17.

As further described below, the microorganism flora in the gas dispersion return sludge 36 has been activated by the oxygen (possibly with a trace of ozone) supplied by the gas generator 28. The gas-dispersion return sludge 36 is either completely devoid of, or has an extremely low level of organic pollutants (and other pollutants that can be digested to obtain energy) that can be digested by the activated microorganisms. At such conditions, the microorganisms have been empirically shown to prioritize their digestive function at the expense of their reproductive function, even though the microorganisms have no way to capitalize on the function at that point in time due to a complete or almost complete absence of digestible material. Upon the gas dispersion return sludge 36 being mixed with the wastewater that includes digestible organic pollutants (and other pollutants, such as ammonia, that can be digested to produce energy), the microorganisms that have previously been "starved" of digestible material start exhibiting the digestive function at an elevated rate compared to a suppressed reproductive function, thus turning the majority of the pollutants into water and other byproducts. As described further below with reference to equation (3) and (4), the byproducts of the digestion and the oxygen requirements for

the digestion differ from what the byproducts and the oxygen requirements under normal conditions. The prioritizing of the digestive function over the reproductive function continues within the one or more Aerobic Reaction Vessels 16 until the energy requirements of the microorganism have been satisfied, at which point they can also start exhibiting the reproductive function to produce sludge 21 if any organic pollutants (or other absorbable pollutants) remain in the mixed liquor 17 undigested. Upon complete or substantially complete consumption of the digestible or absorbable pollutants within the mixed liquor, at least a portion of the microorganisms (such as the majority or all of the microorganisms within the mixed liquor) enter a dormant state (in which biochemical processes within the microorganisms are substantially slowed down or halted) due to a lack of further digestible material.

Upon completion of a hold time, a time that can be experimentally determined to be adequate for the completion of the digestion (and possibly consumption through the reproductive function) of the organic pollutants (and other pollutants, such as ammonia, on which the functions can be exhibited) in the mixed liquor 17, the mixed liquor 17 is pumped from the Aerobic Reaction Vessels 16 to the Sludge Sedimentation Vessel 18, where the mixed liquor 17 is separated into a supernatant and a sludge 21. The sludge 21 is collected in Sludge Storage Vessel 20, and as described further below, is returned to Aerobic Reaction Vessel 16 in the form of gas-dispersion return sludge 36, for cyclical reuse. In a further embodiment, the system 10 can omit the Sludge Storage Vessel 20, and the exit side of Sludge Sedimentation Vessel 18 and the Mixer/Distributor 15 are connected to the Return Sludge Pipeway 26.

The supernatant is pumped out from the Sludge Sedimentation Vessel 18 as Treated Water 72, which in turn is pumped into a Treated Water Processing Unit 73, where the water is further processed. Such processing can include sterilization, such as described in U.S. Pat. No. 10,287,194, filed issued May 14, 2019, to Ohki et al., the disclosure of which is incorporated by reference, though other ways to sterilize the Treated Water 73 is possible. Other processing can be done at the unit 73. While the unit 73 is shown as a single physical structure with reference to FIGS. 1A and 1B, the unit 73 could also be made up of multiple, spatially separated components. The water processed by unit 73 is expelled from the system 10 as Purified Water 74, now being suitable for use, such as drinking.

Additionally, the excess sludge is pumped out of the wastewater management system as Excess Sludge 27.

At least some portion of the sludge from Sludge Sedimentation Vessel 18 passes through the Return Sludge Pipeway 26 in the form of Return Sludge 35, entering the Atomizer 24 (in the embodiment of the system 10 shown with reference to FIG. 1A) or the Cavitation Pump 22 (in the embodiment of the system 10 shown with reference to FIG. 1B). Before the entrance point of the Atomizer Pump 24 or the Cavitation Pump 22, pure oxygen gas or pure oxygen gas with trace amounts of ozone are mixed into Return Sludge 35, forming gas-dispersion return sludge 36.

The creation of the gas-dispersion return sludge 36 is made possible by the use of the Atomizer 24 (in the embodiment of the system 10 shown with reference to FIG. 1A) or the Cavitation Pump 22 (in the embodiment of the system 10 shown with reference to FIG. 1B). In particular, a gas-liquid mixture (a mixture of the at least one reactive gas 37 with the Return Sludge 35) is formed within the Return Sludge Pipeway 26 and is pumped to the Atomizer 24 (in the embodiment of the system 10 shown with reference

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to FIG. 1A), which has the function of churning and mixing the aforementioned gas-liquid under high pressure (approximately 0.0981-5.394 MPa (1-55 kg/cm²)), then employing either cavitation or 20-12,000 kHz ultrasound respectively or both simultaneously to induce ultra-fine bubbles in the gas-liquid mixture of diameter from 1 nm-30,000 nm, further causing oxygen radicalization and hydroxyl radicalization. A portion of the ultra-fine bubbles dissolve within the gas-dispersion return sludge, raising the level of the dissolved oxygen to the critical threshold of at least 10 mg/l (with the concentration of ozone, if ozone is utilized, being 0.01-0.5 mg/l), and a portion is stored with the sludge 36 as ultra-fine bubbles. Thus, the Atomizer 24 can instantaneously render the desired quantity of reactive gas 37 into ultra-fine bubbles, rapidly dissolving a portion of the reactive gas, then disperse, immobilize and store the excess in a liquid in the form of ultra-fine bubbles. Increasing the level of the dissolved oxygen to the critical threshold (at least 10 mg/l) has been empirically shown to activate the microorganisms within the sludge 36, removing them from the dormant state. While the microorganism are activated at this point and are ready to digest organic and other digestible pollutants (and are prioritizing their digestive function over their reproductive function), the level of digestible pollutants (or other digestible organic materials) within the gas-dispersion return sludge 36 is either at zero or close to zero (at a level insufficient to satiate the prioritized digestive function of the microorganism), and thus the microorganisms are forced into a state of extreme starvation.

In one embodiment, the Atomizer 24 can be the OHR Mixer sold by OHR Laboratory Corporation of 536-1, Noda, Irumashi, Saitama 358-0054 Japan. In a further embodiment, other Atomizers 24 can be used.

Similarly, in the embodiment of the system 10 shown with reference to FIG. 1B, instead of entering the Atomizer 24, the mixture of the return sludge 35 and the at least one reactive gas 37 enters a Cavitation Pump 22. Cavitation is the formation of vapor cavities in a liquid. In pumps, cavitation is caused by an impeller of the pump moving through a liquid, with low-pressure areas being formed as the liquid accelerates and moves past the blades, causing the liquid to vaporize and form small bubbles of gas. While cavitation in most cases is undesirable as cavitation is damaging to the components of the pump, the Cavitation Pump 22, while subject to increased wear due to cavitation, takes advantage of the cavitation effect to help dissolve at least one reactive gas 37 within the return sludge to create gas-dispersion return sludge 36. In particular, the rotation of the impeller of the Cavitation Pump 22 is fast enough to slice the formed bubbles into multiple smaller bubbles, thus forming ultra-fine bubbles of the 1 nm-30,000 nm diameter. The Cavitation Pump 22 operates under a high pressure, which facilitates the dissolution of the at least one reactive gas 37 within the return sludge 35. In one embodiment, the pressure inside the pump 22 is between 0.0981 MPa and 5.394 MPa, though other values of pressure are also possible.

In a still further embodiment, the system 10 could include both the Cavitation Pump 22 and the Atomizer 24, with both the Cavitation Pump 22 and the Atomizer 24 contributing to the creation of the ultra-fine bubbles and creation of the gas-dispersion return sludge 36. The gas-dispersion return sludge 36 is returned solely to Mixer/Distributor 15 and the one or more Aerobic Reaction Vessels 16, and is not returned to Sedimentation Vessel 12, Adjustment Vessel 14, or Sludge Sedimentation Vessel 18. The returned quantity of gas-dispersion return sludge 36 is unitarily controlled and uni-

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tarly managed to maximize the sum total reduction of carbon dioxide, the reduction in treatment costs, and the reduction in energy usage of the entire wastewater treatment system. Because the wastewater treatment capacity of one or more Aerobic Reaction Vessels 16 is dramatically increased, the Aerobic Reaction Vessel 16 can be made very small.

Due to processing by the Atomizer 24 or the Cavitation Pump 22 and being subsequently provided to the one or more Aerobic Reaction Vessels 16, the microbe flora within gas-dispersion return sludge 36 becomes activated by the oxygen (and possibly ozone) that the microbes receive. For example, when the gas-dispersion return sludge enters Aerobic Reaction Vessel 16 (or if the Mixer/Distributor 15 is employed, the Mixer 15), the activated microbes have been brought to a state of extreme near-starvation, so that between their reproductive function and their digestive function, they autonomously prioritize digestion over reproduction. Thus, the provision of the high level of oxygen (or oxygen with the trace of ozone) allows the operator to individually control and manage the reproductive and digestive functions of the microbe colonies, prioritizing the digestive function of the microbes over the reproductive function. In the present application, the term "reproductive function" of the microbe flora is defined as the function by which the microbes absorb as food organic matter contained in wastewater 11, the microbes grow, and then reproduce, such that the organic matter becomes isolated as a mass of matter and microbes, thus purifying the wastewater. The "digestive function" of the microbe colonies is defined as the function by which the microbes absorb as food organic matter contained in wastewater, then break it down and digest it to gain energy which sustains their activity and their life processes.

The high activation of the microorganisms is particularly pronounced when the concentration of oxygen within the Aerobic Reaction Vessel 16 reaches a particular critical threshold, with the presence within the mixed liquor 17 of at least 10% of volume of the gas-dispersion return sludge 36 that has at least 10 mg/l of dissolved oxygen (a level that can be achieved through the use of the Atomizer 24 or the Cavitation Pump 22). Once that threshold is reached, 1DO (mg/l) is enough for breaking down more pollutants than what would be included in 1BOD or 1COD when the dissolved oxygen within the Aerobic Reaction Vessel 16 is at a lower level. At these conditions, the amount of pollutants degraded by the microorganisms within the Aerobic Reaction Vessel 16 has experimentally been shown to exceed 20 times the amount of pollutant degraded by the same microorganisms than when the concentration of oxygen is at levels close to those occurring in nature (2-4 mg/l). At a differing concentration of oxygen, the critical threshold of the gas-dispersion return sludge 36 that needs to be added to the Aerobic Reaction Vessel 16 would change proportionally. The knowledge of this effect can be used to calculate with great precision an amount of the gas-dispersion return sludge 36 that is necessary for purification of a particular amount of wastewater 11, allowing to reduce the amount of unnecessary Excess Sludge 27 produced. Likewise, knowing the volumes of the gas-dispersion return sludge 36 and the volume of wastewater 11 that need to be handled can allow to properly size the components of the system 10, reducing waste and cost of creating the system 10.

As the formation of ultra-fine bubbles plays an important role in the activation of the microorganisms, additional explanation is provided regarding the formation and use of the bubbles below. Regarding the slowing effect on the

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velocity at which bubbles rise within a liquid which can be achieved by producing bubbles which are ultra-fine, bubbles with diameter of around 30 μm rise within a liquid at approximately 1 m/hr, and at a diameter of around 1 μm they rise at less than 0.005 m/hr (Stokes' Law for Spherical Bubbles). With this range of velocity, bubbles remain within the liquid for long enough that they can immediately and at the required position replace dissolved oxygen which has been consumed by the biochemical reaction with the pollutant substances in the wastewater to be treated, and furthermore, because the bubbles can be dispersed in ultra-fine bubble form, uniformly and in great quantity, and therefore in the same places where oxygen has been consumed, a bubble storage function is also achieved.

In this way, the desired reactive gases including oxygen or oxygen and ozone can be supplied and stored with extremely long duration, with neither surplus nor shortage, thereby shortening and stimulating the biochemical reaction, and also allowing that the supply within the time period required to carry out the biochemical reaction need not necessarily be continuous but can be intermittent.

As mentioned above, the Atomizer **24** or the Cavitation Pump **22** is employed to disperse gas into liquid in the form of ultra-fine bubbles. To render bubbles to an ultra-fine size and blend the ultra-fine bubbles into liquid, mechanical agitation and cutting are insufficient to achieve the nano level, and only when the velocity of the two-phase flow of the vapor-liquid is increased through pressurization, and a synergistic effect with the vortex churning of the liquid is generated using cavitation and ultrasound, that the bubbles are broken down to ultra-fine state and blended into the mixture as ultra-fine bubbles. For the gas to be dissolved and remain in dissolved state, pressure conditions are of key importance, and higher pressures are known to be more advantageous. Taking all these factors into account, the range of pressure chosen for the Atomizer **24** or the Cavitation Pump **22** is from 0.0981 MPa-5.394 MPa (1-55 kg/cm²).

While in a simple return process for activated sludge (with zero addition of reactive gas), operating in low pressure ranges helps to avoid destroying the microorganisms which exist in the activated sludge, maintaining the pressure low no longer makes sense when reactive gas is added to the sludge. The reason to strive for the highest pressure that can be practically achieved (approximately 5.5 MPa), is to effectively utilize, in the oxidation and decomposition process of sludge employing reactive gas including high density ozone, a synergistic oxidation and decomposition effect between the actions of cavitation and ultrasound, which under high pressure cause the oxidation and breakdown of ozone itself, and the functioning of O radicals and OH radicals. With the large-capacity wastewater treatment employing the activated sludge method **40** and the system **10**, care has been taken to choose frequencies of ultrasound which can be used easily and economically, and so at low pressure ranges a frequency of 20 kHz was chosen, and for high pressure ranges (approximately 5.5 MPa) a frequency of 12,000 kHz was chosen. In a further embodiment, other frequencies in the 20 kHz-12,000 kHz range are possible.

For the Oxygen and Ozone Generator **28**, an ozone generator or similar may be used to regulate the supply of oxygen and the generation of ozone. For example, by employing an ozone-generating element comprising an electrode mounted to a boss fashioned from a dielectric substance, and a high-frequency high-voltage power source which applies a high-frequency alternating current to the ozone-generating element while supplying an oxygen-rich

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gas to the ozone-generating element, and adjusting the quantity of ozone generated by using a regulator to control the voltage and/or the frequency of the power source, it becomes possible to effect an oxygen/ozone cycle generator which regulates the amount of oxygen and ozone supplied, to cope with fluctuations in the quality and load of sewage for wastewater treatment due to morning, daytime or nighttime, or due to either dry weather or rainy weather, or to cope with processes based mainly on the supply of oxygen and with processes based mainly on oxidation and decomposition by ozone. For the reactive gas including oxygen to be supplied, oxygen-enriched air or pure oxygen are both acceptable. The supplied gas may also be pumped as is, with zero ozone generation. Of course, the operation of the oxygen/ozone cycle generator may also be suspended. Further, while the Oxygen and Ozone generator **28** is represented as a single unit, in a further embodiment, the system **10** could include multiple generators **28**, one generator **28** providing ozone and another generator providing oxygen, with the gases provided by both generators being provided into the Return Sludge Pathway **26** to be mixed with the return sludge **35**.

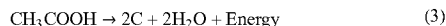
In the activated sludge process, the microorganisms which effect the biochemical reaction are returned to the wastewater intake side with a portion of the sludge (return sludge) such that the microorganisms are utilized cyclically. If the wastewater **11** to be treated includes high densities of organic substances, and accelerating the microbial biochemical reaction is therefore necessary, then maximizing the quantity of oxygen dissolved in the wastewater **11** or replenishing dissolved oxygen rapidly according to the amount of dissolved oxygen which is consumed is desirable. The system **10** performs favorably in this respect, employing the Atomizer **24** or the Cavitation Pump **22**, to infuse with the required amount of oxygen gas (or oxygen with the trace of ozone) the water which carries the return sludge back to Aerobic Reaction Vessel **16**. The microbial biochemical reaction is accelerated dramatically due to the Atomizer **24** or the Cavitation Pump **22** supplying a plentiful amount of oxygen (or oxygen with a trace of ozone) in a dissolved state and in the form of ultra-fine bubbles in an extremely short time.

Because ultra-fine bubbles, as previously described, require a very long time to float to the surface of Aerobic Reaction Vessel **16**, during the time which it takes them to float to the surface of Aerobic Reaction Vessel **16**, the ultra-fine bubbles in Aerobic Reaction Vessel **16** are dispersed and stored in the form of ultra-fine bubbles, and continuously replenish the dissolved oxygen. By maintaining a high quantity of dissolved oxygen in Aerobic Reaction Vessel **16**, significant acceleration of the microbial biochemical reaction becomes possible. Due to the effect of the microbial biochemical reaction within Aerobic Reaction Vessel **16**, a portion of the organic matter in the wastewater is digested, releasing carbon dioxide and water, and a portion of the organic matter is consumed by activated sludge microorganisms; the microorganisms multiply, and the activated sludge is generated. In this case, by adding not only oxygen to the wastewater, but by optionally also adding and employing trace amounts (e.g. up to 0.01-0.5 mg/l=ppm) of ozone, greater activation of the microorganisms which carry out the microbial biochemical reaction becomes possible.

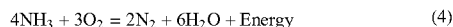
Further, under the conditions that are created in the system **10** of FIG. 1, the digestive function exhibited by the microorganisms while in the aerobic reaction vessels occurs via different biochemical pathways (and consequently via dif-

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ferent chemical reactions) than under natural conditions. In particular, in addition to water, at least some of the biochemical pathways result in production of inert compounds, such as amorphous carbon or diatomic nitrogen (N₂). For example, under the conditions created in the system 10, digestion of acetic acid proceeds in accordance with the following equation:



Compared to equation (1) given above, the chemical reaction of equation (3) that takes place under the conditions in the Aerobic Reaction Vessels 16 created in system 10 requires no external oxygen, thus allowing the reaction to proceed until the energy requirements of the microorganisms satisfied, making exhibition of the reproductive function possible after that point. Further, unlike the reaction given in equation (1), the reaction does not produce carbon dioxide, but instead produces inert amorphous carbon that can be easily removed from the mixed liquor 17. While this reaction does not use oxygen, the same microorganisms in the mixed liquor 17 can also perform aerobic digestion of other compounds. For example, digestion of ammonia takes place via a different chemical reaction than what is given in equation (2), but is still aerobic. Thus, under the conditions created in the system 10, in the Aerobic Reactions Vessels 16, the digestion of ammonia takes place in accordance with the following equation:



This reaction of equation (4) uses only 3 molecules of diatomic oxygen compared to seven molecules used in the reaction of equation (2)), allowing the oxygen within the ultra-fine bubbles within the mixed liquor 17 to sustain the reaction longer. Further, instead of the nitrogen dioxide produced in equation (2), the reaction of equation (4) produces diatomic nitrogen, which does not pose the same environmental concerns as nitrogen dioxide. Similar reactions can also take place with compounds that include phosphorous (such as phospholipids) and compounds that include sulfur. Likewise, similar reactions can take place with other biodegradable materials including cellulose and other carbohydrates, fats, and oils, and different kinds of protein. Overall, under the conditions created in the Aerobic Reaction Vessels 16, the microorganisms prioritize (if not switch entirely) using the biochemical reaction pathways for digesting multiple, if not all, pollutants that consume less oxygen and that produce different end products than the biochemical pathways for digesting the same pollutants employed under conditions found in nature. Further, at least some of the biochemical pathways utilizing less oxygen also may occur faster than the pathways for degrading the same pollutants that utilize more oxygen.

In one embodiment, the components of the system 10 described above can be controlled independently of each other. In a further embodiment, the system 10 includes a Controller 39 that is interfaced, such as via a wired or a wireless connections, to at least the Sludge Sedimentation Vessel 20, the Oxygen and Ozone generator 28, and the Atomizer 24 (in the embodiment of the system 10 shown with reference to FIG. 1A) or the Cavitation Pump 22 (in the embodiment of the system 10 shown with reference to FIG.

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1B). The Controller 39 can also be similarly interfaced to other components of the system 39. The Controller 39 can receive from the operator of the system 10 the amount of gas-dispersion return sludge 36 that is to be delivered to the Aerobic Reaction Vessel 16 and control the Sludge Sedimentation Vessel, the Oxygen and Ozone Generator 28, the Cavitation Pump 22 or the Atomizer 24 to deliver the desired amount of the gas-dispersion return sludge 36. Alternatively, the controller 39 can receive from the operator a characteristic of the wastewater treatment, such as a degree of the wastewater treatment desired by the operator, a wastewater treatment time desired by the operator, and a desired wastewater treatment capacity, and determine the amount of the gas-dispersion return sludge 36 to be delivered to the Aerobic Reaction Vessel 16 to achieve the desired characteristic. The determined amount can then be delivered under the control of the Controller 39. The Controller 39 can be a computing device, such as a personal computer, a smartphone, a laptop, a tablet, though other kinds of computing devices are possible. The Controller 39 can include components conventionally found in general purpose programmable computing devices, such as a central processing unit, memory, input/output ports, network interfaces, and non-volatile storage, although other components are possible. The central processing unit can implement computer-executable code which can be implemented as modules. The modules can be implemented as a computer program or procedure written as source code in a conventional programming language and presented for execution by the central processing unit as object or byte code. Alternatively, the modules could also be implemented in hardware, either as integrated circuitry or burned into read-only memory components. The various implementations of the source code and object and byte codes can be held on a computer-readable storage medium, such as a floppy disk, hard drive, digital video disk (DVD), random access memory (RAM), read-only memory (ROM) and similar storage mediums. Other types of modules and module functions are possible, as well as other physical hardware components.

The Controller 39 can be controlled by the operator on-site or remotely. For example, the Controller 39 can be interfaced to an Internetwork, such as the Internet or a cellular network, and a user device (such as a smartphone though other user devices are possible) also interfaced allows to command the Controller 39 remotely, and provides remote control of the system 10 to the operator.

Other embodiments of the Controller 39 are also possible.

When the system 10 has not been run recently, there may not always gas-dispersion return sludge 36 available to be added to the Aerobic Reaction Vessel 36 and provide the aerobic microorganisms necessary to conduct the aerobic reaction to the Aerobic Reaction Vessel. In such a situation, the system 10 may utilize seed sludge-sludge 21 that is input into the system 10, such as into the Return Sludge Pipeway 26, from an external source, such as another wastewater treatment system, though other external sources are possible. By being processed by the Atomizer 24 (or the Cavitation Pump 22) and the Oxygen and Ozone Generator 28, the seed sludge is turned into the gas-dispersion return sludge 36 and can then be provided to the Aerobic Reaction Vessel 16 to be used for the treatment of the wastewater 11. As the microorganisms present in activated sludge 21 differ significantly based on the geographic origin of the wastewater 11 from which the sludge 21 is created, the seed sludge introduced into the system 10 is selected based on the geographic location of the wastewater from 21 from the seed sludge originates. Preferably, the seed sludge is from the same or

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proximate geographic location as the wastewater **11** being processed by the system **10** to avoid an introduction of exogenous microorganisms that can negatively impact the aerobic reaction.

The system **10** can be created from most existing water treatment facilities by retrofitting certain portions of the system **10** onto existing equipment. In particular, the Atomizer **24** or the Cavitation Pump **22**, and the oxygen and ozone generator **28** can be retrofitted into an existing wastewater treatment plant, allowing for widespread use of the system **10** and the method described in this application.

As described above, providing the gas-dispersion return sludge **36** allows the operator to exercise increased control over the wastewater purification. In particular, the ratio of the volume of gas-dispersion return sludge **36** to the volume of the wastewater **11** to be treated (and consequently the amount of consumable pollutants available to the microorganisms in the gas-dispersion return sludge **36**) determines whether the microorganisms exhibit exclusively the digestive function, or upon acquiring the necessary energy, in the presence of additional pollutants and oxygen, can proceed to also exhibit the reproductive function and produce the sludge **21**. FIGS. 2A-2B are flow diagrams showing a method **40** for wastewater treatment through controlled microorganism activation in accordance with one embodiment. The method can be implemented using the system **10** of FIG. 1A or 1B. Optionally, if no gas-dispersion return sludge **36** is present in the Aerobic Reaction Vessel at the start of the execution of the method **40**, seed sludge is added to the system **10**, is converted into gas-dispersion return sludge **36**, and is provided into one or more of the Aerobic Reaction Vessels, as described above with reference to FIGS. 1A-1B (step **41**). The load of wastewater **11** to be treated is determined and the amount of gas-dispersion return sludge **36** to be delivered to the Aerobic Reaction Vessel **16** is determined (step **42**). The determination of the amount of the gas-dispersion return sludge **36** can be done based on the load as well as other desired characteristics of the wastewater treatment, such as the degree of the purification and the speed of the treatment, though other characteristics are possible. Another one of the characteristics is whether the goals of the wastewater treatments include only digestion of the pollutants (organic and other degradable pollutants such as ammonia) via the digestive function or whether some creation of additional sludge **21** is desired. Thus, if the goal of the wastewater treatment is pure digestion of the organic pollutants (and other pollutants that can be digested to produce energy, such as ammonia) with creation of minimum to no sludge **21**, the ratio of the volume of the gas-dispersion return sludge **36** to the volume of the wastewater **11** is going to be greater than if creation of at least some sludge **21** is desirable. However, as some amount of the sludge **21** is necessary for subsequent purification cycles, the ratio of the volume of the gas-dispersion return sludge **36** to the volume of the wastewater **11** can be decreased, allowing the microorganism to exhibit their reproductive function following the satiation of the digestive function. The levels of pollutants (organic and other digestible pollutants) within a wastewater **11** can vary with time even in the same location. These levels can also vary based on the source of the wastewater **11** (and hence the geographic location from where the wastewater **11** originates). Similarly, the digestive and reproductive abilities of the microorganism flora vary depending on which strains of the microorganisms that make up the flora in the sludge **36**, with different strains being present in different geographic regions. Further, the exact levels of dissolved reactive gas

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within the gas-dispersion return sludge **36** can affect the digestive ability of the microorganisms within the sludge **36** and thus the amount of the sludge **36** required to achieve purification goals. Due to the variabilities associated with the different geographic locations and levels of reactive gases, the optimum amount of sludge for a particular purpose can be determined experimentally, as further described below. For the purpose of eliminating as much as of the organic pollutants as possible while producing minimum amount of sludge, the ratio in the mixed liquor **17** of volume of the gas-dispersion return sludge **36** (having at least 10 mg/l of dissolved oxygen) to the volume of the wastewater **11** of at least 10% has generally proved sufficient.

Optionally, if there is an opportunity for physically setting up or changing the set-up of the equipment used for processing of the wastewater **11** (such as the system of FIGS. 1A-1B), the parameters of the equipment necessary for processing a particular wastewater load, such as the size (though other parameters are also possible) can be determined and optionally implemented in accordance with the determination (step **43**).

The wastewater **11** enters Sedimentation Vessel **12** where grit and other inorganic solids settle and are separated out (step **44**). Subsequently, the wastewater **11** enters Adjustment Vessel **14** where the load and conditions of raw sewage **11** are adjusted and solid organic material is solubilized by anaerobic microorganisms (step **45**).

Next, the wastewater **11** flows, possibly via a mixer/distributor **15** if one or more of them are part of the system **10**, into one or more Aerobic Reaction Vessels **16**, where the wastewater **11** (raw sewage) is added to gas-dispersion return sludge **36** and blended to form mixed liquor **17** (if the mixer/distributor is involved, the mixed liquor **17** forms within the mixer-distributor **15** and is provided to the one or more Aerobic Reaction Vessels **16** where the majority of the biochemical (aerobic and other kinds of digestion) consumption of organic pollutants (and other degradable pollutants) takes place) (step **46**). From there, if any aeration by bubbling using air as a reactive gas was previously performed, such aeration becomes unnecessary due to the reactive gas contained in gas-dispersion return sludge **36**. Dissolved oxygen (DO) (possibly with traces of ozone) is supplied to the Vessels **16** by the gas-dispersion return sludge **36** and organic solids left undissolved after step **45** are oxidized; at the same time biochemical treatment by aerobic microbe flora occurs, with the organic (and other digestible) pollutant substances dissolved in the wastewater **11** being digested to water (H₂O) and other byproducts, such as inert compounds such as amorphous carbon or diatomic nitrogen (N₂), which can be easily removed. If the digestive function of the microorganisms has been satisfied and exhibition of the reproductive function is possible, some of the pollutants can be immobilized as additional sludge **21** via the exhibition by the microorganisms of their reproductive function (step **47**). Following the completion of the consumption of the organic pollutants via the digestive and possibly the reproductive function, the microorganisms enter the dormant state.

Through the action of the Atomizer **24** or the Cavitation Pump **22**, and the Oxygen and Ozone Generator **28**, the quantity of dissolved oxygen in gas-dispersion return sludge **36** increases to a critical level (at least 10 mg/l) that activates the microbe flora within the gas-dispersion return sludge **36** from the dormant state. While the microorganisms are activated at this point and are ready to digest the organic and other pollutants (and are prioritizing the digestive function over the reproductive function), the level of the organic

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pollutants (and other pollutants which can be digested to extract energy) is either at zero (with the pollutants having previously been entirely consumed at the one or more Aerobic Reaction Vessels) or close to zero (at a level insufficient to satiate the prioritized digestive function of the microorganisms), and thus the microorganisms are forced into an extreme state of near-starvation. The activated microbe flora which exists within gas-dispersion return sludge 36 is supplied to Mixer/Distributor 15, if present in the system 10, and the activated microbe flora is blended with wastewater 11 by the Mixer/Distributor 15 to form the mixed liquor 17, which is supplied to the one or more Aerobic Reaction Vessels 16 (or if the system 10 does not include the Mixer/Distributor 15, the mixed liquor 17 is formed within the Aerobic Reaction Vessel 16). Within the Aerobic Reaction Vessels, the microorganisms continue to autonomously prioritize their digestive function over their reproductive function. The efficiency ratio of oxygen (DO) utilization in the digestive function of the microbe flora becomes extremely heightened. As a result, the activated microbe flora in the one or more Aerobic Reaction Vessels 16, becomes capable of purifying far more organic matter per 1DO than the typically defined quantity for the activated sludge method of 1BOD or 1COD.

Next, the mixed liquor 17 progresses to Sedimentation Vessel 18, settles inside Sedimentation Vessel 18 and is separated into sludge 21 and supernatant (step 48). The settled sludge 21 is collected in Sludge Storage Vessel 20 (step 49), and is returned to Aerobic Reaction Vessel 16 in the form of gas-dispersion return sludge 36, for cyclical reuse (step 50), as further described below with reference to FIG. 3.

Excess sludge 27 is expelled from the Sludge Storage Vessel 20 and from the system 10 (step 51). Supernatant is also removed as treated water 72, and undergoes further processing at the Treated Water Processing Unit 73 before being expelled from the system as Purified Water 74 (step 52). The sterilization can be performed as described in U.S. Pat. No. 10,287,194, issued May 14, 2019, to Ohki et al., the disclosure of which is incorporated by reference, though other ways to perform the sterilization are possible.

Optionally, the amount of the sludge 21 that is collected in the Sludge Storage Vessel can be measured (such as by weighing the sludge 21), with the results of the measurement being used to adjust the ratio of gas-dispersion return sludge 36 to wastewater 11 used to form the mixed liquor 17 in subsequent repetitions of the method 40 (step 53). For example, if more than a desired amount of sludge 21 was measured, the ratio of the amount of the gas-dispersion return sludge 36 to the amount of the wastewater 11 used to form the mixed liquor 17 can be increased in subsequent runs of the method 40 (thus reducing the overall amount of digestible pollutants available to the microorganisms and increasing the probability that all of the degradable pollutants will be digested by the microorganisms via the digestive function). Alternatively, if not enough sludge 21 was created, the ratio of the amount of the gas-dispersion return sludge 36 to the amount of the wastewater 11 used to form the mixed liquor 17 can be decreased in subsequent runs of the method 40 (thus increasing the amount of the degradable pollutants available to the microorganisms, which, can be used for reproduction of the microorganism following the satiation of the digestive function).

If more wastewater 11 to be treated remains (step 54), whether the amount of solid pollutants, organic and inorganic, in the next batch of wastewater 11 to be treated requires action via execution of steps 44 and 45 is deter-

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mined (step 55). The determination can be made by comparing the level of the solids to one or more thresholds, though other kinds of determinations are possible. If the level requires action (step 55), the method returns to step 44. If the level does not require action (55), the method 40 returns to step 46. If no more wastewater 11 remains to be processed (step 52), the method 40 ends.

Providing the gas-dispersion return sludge 36 solely into the one or more Aerobic Reaction Vessels 16, possibly via the Mixer/Distributor 15, allows to achieve an optimum quantity of the aerobic microorganisms within the Aerobic Reaction Vessels 16. FIG. 3 is a flow diagram showing a routine for forming gas-dispersion return sludge and returning the gas-dispersion return sludge 36 to the one or more aerobic reaction vessels 16 for use in the method of FIG. 2A-2B in accordance with one embodiment.

Reactive gas, pure oxygen or oxygen containing trace amounts of ozone, is generated by the Oxygen and Ozone Generator 28 (step 61). As described above, either the Atomizer 24 or the Cavitation Pump 22 is installed along the Return Sludge Pipeway 26. Once the reactive gas is introduced into the return sludge 35, the return sludge 35 is converted into a gas-liquid mixed liquor (step 62). When this gas-liquid mixed liquor (sludge) passes through the Atomizer 24 or the Cavitation Pump 22, the reactive gas within the gas-liquid mixed liquor (sludge) is instantaneously rendered into ultra-fine bubbles (bubble diameter less than 30 μ m, ideally bubble diameter less than 1 μ m) and a portion of it is instantly dissolved (step 63). With this, a super-saturated DO value of 10-40 mg/l is realized (0.01-0.5 mg/L of ozone if ozone is also generated), and the remaining gas is dispersed, immobilized and stored within the sludge in an ultra-fine bubble state, providing a way to replenish the supply of the dissolved reactive gases and continue the digestion of the degradable pollutants.

This gas-dispersion return sludge 36 containing reactive gas is supplied by the Atomizer 24 or the Cavitation Pump 22 only to the one or more Aerobic Reaction Vessels 16 (possibly via the Mixer/Distributor 15), where the gas-dispersion return sludge 36 is formed, along with the wastewater 11, part of the mixed liquor 17 (step 64), ending the routine 60.

As mentioned above, upon addition of the gas-dispersion return sludge to one or more of the Aerobic Reaction Vessels 16 any bubbling in the Aerobic Reaction Vessel 16 can be ceased. Or, in cases where the bubbling is required to prevent the settling of sludge, bubbling aeration can be minimal and may be conducted intermittently and for short periods of time.

While the invention has been particularly shown and described as referenced to the embodiments thereof, those skilled in the art will understand that the foregoing and other changes in form and detail may be made therein without departing from the spirit and scope of the invention.

What is claimed is:

1. A pump-based system for wastewater treatment through microorganism biochemical pathway optimization, comprising:

a gas generator configured to provide at least one reactive gas into a sludge comprising microorganisms having a digestive function and capable of digesting at least one pollutant via at least two biochemical pathways, one of the pathways for digesting that pollutant producing inert byproduct and another one of the pathways producing a further byproduct other than the inert byproduct, wherein the sludge is substantially free of the pollutants;

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- a controller interfaced to the gas generator and a pump and configured to receive user input comprising a desired degree of the digestion of the at least one pollutant in wastewater via the digestive function and to determine an amount of a gas-dispersion return sludge to be combined with an amount of the wastewater to form a mixed liquor based on the user input; and
- the pump configured to:
- form the gas-dispersion return sludge by rendering the at least one reactive gas into ultra-fine bubbles within the sludge, wherein a portion of the ultra-fine bubbles dissolves within the sludge; and
 - provide under a control of the controller the determined amount of the gas-dispersion return sludge for combining with the amount of the wastewater that comprises the at least one pollutant to form the mixed liquor, wherein, upon encountering the pollutant within the wastewater, the activated microorganisms digest the at least one pollutant by prioritizing the pathway that produces the inert byproduct over the pathway that produces the other byproduct.
2. A system according to claim 1, wherein the pollutant comprises one of acetic acid and ammonia.
 3. A system according to claim 1, wherein the inert byproduct comprises one of diatomic nitrogen and amorphous carbon.
 4. A system according to claim 1, wherein the pathway that produces the inert byproduct consumes less of the at least one reactive gas than the pathway that produces the other byproduct.
 5. A system according to claim 1, wherein the at least one reactive gas comprises oxygen, at least some of the microorganisms in the provided sludge are dormant, and the at least one reactive gas dissolution activates the dormant microorganisms.
 6. A system according to claim 5, wherein a concentration of the oxygen within the gas-dispersion return sludge is at least 10 mg/l.
 7. A system according to claim 1, wherein the controller is interfaced to the gas generator and one the pump via one of wired and wireless connections.
 8. A system according to claim 1, wherein the inert byproduct is removed from the mixed liquor.
 9. A system according to claim 1, further comprising a sludge sedimentation vessel in which supernatant is separated from the mixed liquor.
 10. A system according to claim 1, further comprising a treated water processing unit in which the supernatant undergoes further processing before being expelled as purified water.
 11. A pump-based controlled system for wastewater treatment through microorganism biochemical pathway optimization, comprising:
 - a gas generator configured to provide at least one reactive gas into a sludge comprising microorganisms having a digestive function and capable of digesting at least one

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- pollutant via at least two biochemical pathways, one of the pathways for digesting that pollutant producing inert byproduct and another one of the pathways producing a further byproduct other than the inert byproduct, wherein the sludge is substantially free of the pollutants;
- a controller interfaced to the gas generator and a pump and configured to receive user input comprising a desired treatment time of the wastewater and to determine an amount of a gas-dispersion return sludge to be combined with an amount of the wastewater to form a mixed liquor based on the user input; and
- the pump configured to:
- form the gas-dispersion return sludge by rendering the at least one reactive gas into ultra-fine bubbles within the sludge, wherein a portion of the ultra-fine bubbles dissolves within the sludge; and
 - provide under a control of the controller the determined amount of the gas-dispersion return sludge for combining with the amount of the wastewater that comprises the at least one pollutant to form the mixed liquor, wherein, upon encountering the pollutant within the wastewater, the activated microorganisms digest the at least one pollutant by prioritizing the pathway that produces the inert byproduct over the pathway that produces the other byproduct.
12. A system according to claim 11, wherein the pollutant comprises one of acetic acid and ammonia.
 13. A system according to claim 11, wherein the inert byproduct comprises one of diatomic nitrogen and amorphous carbon.
 14. A system according to claim 11, wherein the pathway that produces the inert byproduct consumes less of the at least one reactive gas than the pathway that produces the other byproduct.
 15. A system according to claim 11, wherein the at least one reactive gas comprises oxygen, at least some of the microorganisms in the provided sludge are dormant, and the at least one reactive gas dissolution activates the dormant microorganisms.
 16. A system according to claim 15, wherein a concentration of the oxygen within the gas-dispersion return sludge is at least 10 mg/l.
 17. A system according to claim 11, wherein the controller is interfaced to the gas generator and the pump via one of wired and wireless connections.
 18. A system according to claim 11, wherein the inert byproduct is removed from the mixed liquor.
 19. A system according to claim 11, further comprising a sludge sedimentation vessel in which supernatant is separated from the mixed liquor.
 20. A system according to claim 11, further comprising a treated water processing unit in which the supernatant undergoes further processing before being expelled as purified water.

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